

Civil Engineering Capstone Project

Finite Element Analysis & Terrain Integration of a Three-Story RC Landmark

Conceptual RC Building Design and Structural Analysis using MIDAS Civil

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SketchUp

MIDAS Civil

RC Design

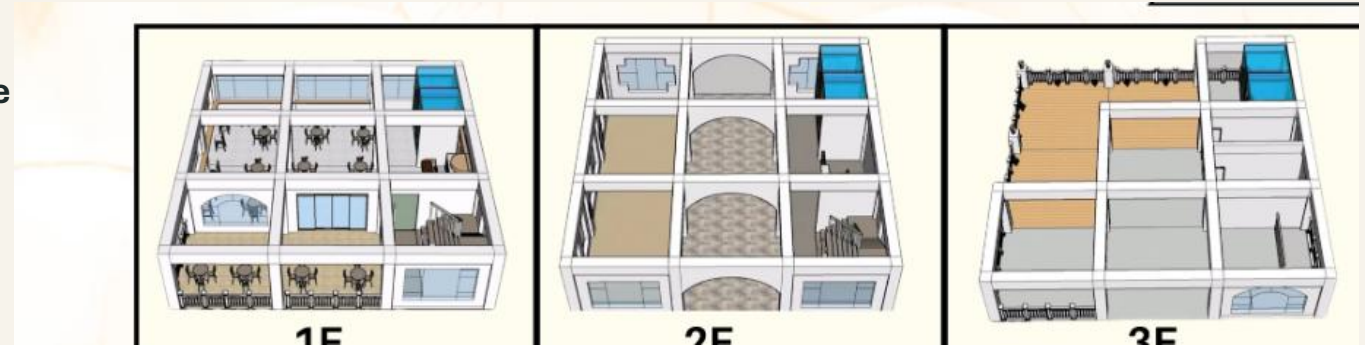


Architectural model

Project Concept, Site & Architectural Program

Design intent

- Create a recognizable three-story building that adds a new visual identity to the campus.
- Use a European-inspired architectural language while integrating landscaped and dining / resting spaces.
- Balance architectural experience with a conceptual reinforced-concrete structural system.



Site-selection logic

- Located near the campus entrance and a major intersection with high pedestrian flow.
- Connects to existing walkways while preserving the lakeside green corridor and lake-facing views.

Three levels were assigned distinct public-space functions within one conceptual RC building

1F — Multipurpose exhibition / public space
2F — Café, dining & discussion area
3F — Viewing platform & sky garden

Structural Design Parameters



Conceptual 3-story RC building

- Plan size: 15 m × 15 m
- Story heights: 4.0 / 3.5 / 3.0 m
- Columns: 50 × 50 cm, 14-D36, D13@50 cm
- Beams: 50 × 60 cm, 4-D25, D10@25 cm
- Concrete compressive strength: 280 kgf/cm²
- Reported allowable foundation load: 706.013 tons-force
- Foundation: 9 isolated footings
- Footing depth: 2.0 m
- Footing size: 0.6 × 0.6 m

Field Survey & Site Reconstruction Workflow



Drone imagery of the lake site

1 UAV survey & RTK control

Capture aerial imagery and obtain RTK control-point coordinates.

2

Metashape alignment

Import aerial images, calibrate with control points, and generate a point cloud.

3

3D reconstruction

Build the site model in Metashape and export an OBJ file.

4

Transmutr conversion

Convert the OBJ model to SKP for transfer into SketchUp.

5

SketchUp refinement

Refine terrain geometry, scale, roads, and the architectural model.

6

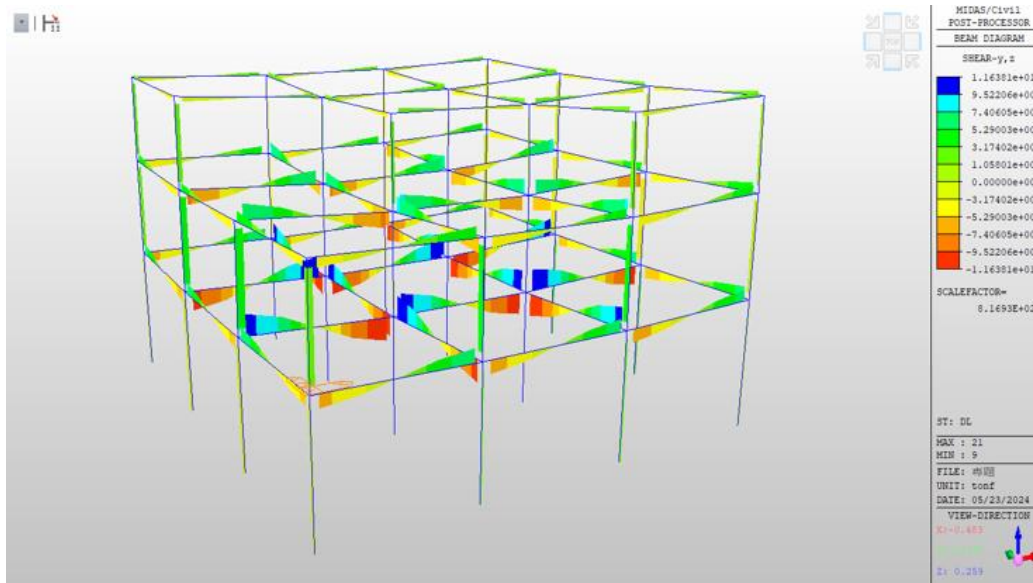
Site integration

Merge the building model with the reconstructed NCHU Lake site and existing paths.

MIDAS Civil Analysis — Internal Force Results

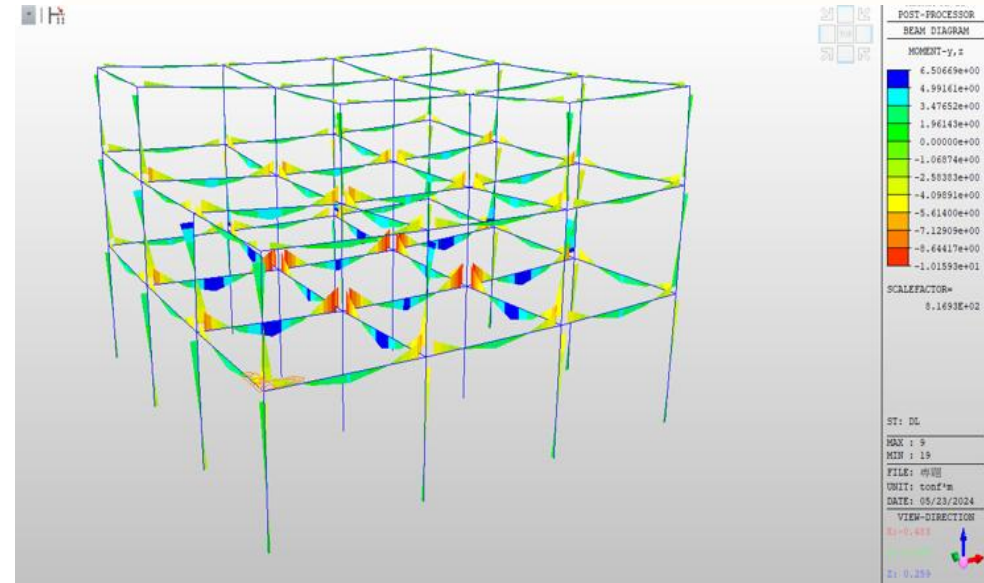
Model setup: RC material and beam/column sections were defined, followed by construction of the 3-story frame model and structural analysis in MIDAS Civil.

Shear force diagram



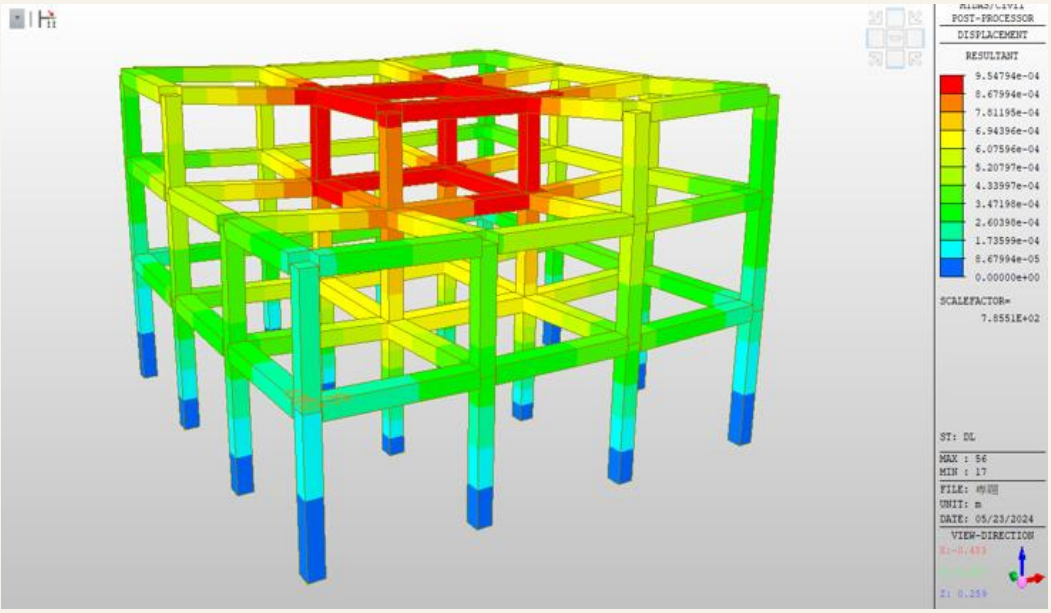
- The displayed MIDAS result windows show ST: DL, indicating the dead-load case in the source screenshots.

Bending moment diagram



- These plots complement the displacement contour by showing internal-force demand rather than deformation magnitude.

MIDAS Civil Analysis — Displacement & Member Checks



MEMB	SECT	SEL	Section	Bc	Hc	fc'	fy	POS	CH
Span			bf	hf		fys			
1			beam			2800.00	I	OK	
1			0.6000	0.5000	24000.0	M	OK		
5.0000			0.0000	0.0000	24000.0	J	OK		
2			beam			2800.00	I	OK	
1			0.6000	0.5000	24000.0	M	OK		
5.0000			0.0000	0.0000	24000.0	J	OK		
3			beam			2800.00	I	OK	
1			0.6000	0.5000	24000.0	M	OK		
5.0000			0.0000	0.0000	24000.0	J	OK		
4			beam			2800.00	I	OK	
1			0.6000	0.5000	24000.0	M	OK		
5.0000			0.0000	0.0000	24000.0	J	OK		
5			beam			2800.00	I	OK	
1			0.6000	0.5000	24000.0	M	OK		
5.0000			0.0000	0.0000	24000.0	J	OK		
6			beam			2800.00	I	OK	
1			0.6000	0.5000	24000.0	M	OK		
5.0000			0.0000	0.0000	24000.0	J	OK		
7			beam			2800.00	I	OK	
1			0.6000	0.5000	24000.0	M	OK		
5.0000			0.0000	0.0000	24000.0	J	OK		
8			beam			2800.00	I	OK	
1			0.6000	0.5000	24000.0	M	OK		
5.0000			0.0000	0.0000	24000.0	J	OK		
25			colume			2800.00	24000.0	OK	
2			4.5000	24000.0	OK				
26			colume			2800.00	24000.0	OK	
2			4.5000	24000.0	OK				
27			colume			2800.00	24000.0	OK	
2			4.5000	24000.0	OK				
28			colume			2800.00	24000.0	OK	
2			4.5000	24000.0	OK				
29			colume			2800.00	24000.0	OK	
2			4.5000	24000.0	OK				
30			colume			2800.00	24000.0	OK	
2			4.5000	24000.0	OK				
31			colume			2800.00	24000.0	OK	
2			4.5000	24000.0	OK				
32			colume			2800.00	24000.0	OK	
2			4.5000	24000.0	OK				
33			colume			2800.00	24000.0	OK	
2			4.5000	24000.0	OK				
34			colume			2800.00	24000.0	OK	
2			4.5000	24000.0	OK				
35			colume			2800.00	24000.0	OK	
2			4.5000	24000.0	OK				
36			colume			2800.00	24000.0	OK	
2			4.5000	24000.0	OK				
37			colume			2800.00	24000.0	OK	

Member-check output

The MIDAS beam and column check tables report CHK = “OK” for the displayed members.

Displacement contour

The model exhibits an overall downward deformation pattern under the selected load case. Contour colors represent displacement magnitude, not load magnitude.

Displacement, internal-force demand, and member checks are separate analysis outputs and should be interpreted independently.